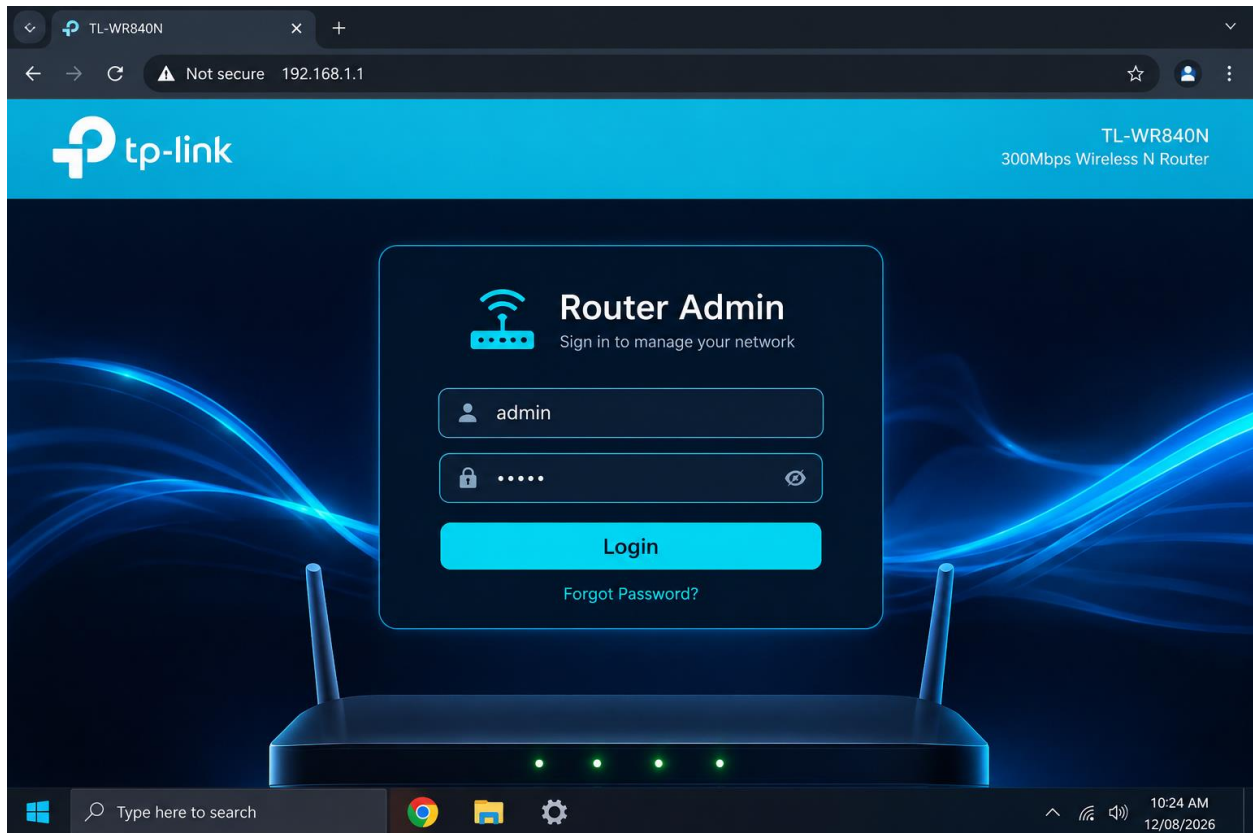


1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.



- I opened the **web browser**.
- I entered the IP address **169.254.206.11** in the address bar.
- The router **login/admin page** is shown in the image.
- I can enter the **username and password** to access the router settings.
- From the Admin Panel, I can manage **Wi-Fi, Internet, security, and other router settings**.

2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

1. Connect your laptop to your Wi-Fi.

2. Press **Windows + R**.

3. Type cmd and press **Enter**.

4. Run:

Ipconfig

5. Under **Wireless LAN adapter Wi-Fi**, note:

- a. **IPv4 Address** → Current IP address

- b. **Subnet Mask** → Subnet mask

- c. **Default Gateway** → Default gateway

Result like this ::

1. **IP Address:** 10.23.121.27

2. **Subnet Mask:** 255.255.255.0

3. **Default Gateway:** 10.23.121.232

3 Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

I checked the current Wi-Fi network settings of my laptop. My router's IP address is in the 10.23.121.x range, with the current gateway being 10.23.121.232.

I then changed the IP configuration from “**Obtain an IP address automatically (DHCP)**” to “**Use the following IP address**” and entered a static IP in the same network range.

The static IP settings I used were:

- **IP Address:** 10.23.121.50
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 10.23.121.232

After applying the static IP settings, I tested the internet connection by opening websites in a browser. **The internet access worked successfully**, which confirmed that the static IP configuration was working.

Finally, after completing the test, I changed the network settings back to “**Obtain an IP address automatically (DHCP)**” so that the laptop can receive its IP address automatically and avoid future network connectivity problems.

4 . In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

Ans ::

I opened my **Wi-Fi router's Admin Panel** and went to the **DHCP settings** section. DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses to devices connected to the Wi-Fi network.

The current DHCP range is:

DHCP Range: 10.23.121.2 – 10.23.121.254

This means the router can automatically assign IP addresses to connected devices within this range.

- ✓ If **two devices are assigned the same IP address**, an **IP address conflict** will occur.
- ✓ The router may not be able to correctly identify the two devices, which can cause **internet connection problems, unstable network access, or difficulty communicating with other devices**.
- ✓ Therefore, every device on the same network should have a **unique IP address**.

